

Enterprise Infrastructure Monitoring & Incident Escalation Platform

Hybrid Cloud Production Architecture & Operations Specification Document

Target Domain: Hybrid Cloud & Edge On-Premises	Platform Version: v2.4 (Production Active)
Core Engine: Python 3.10 / Prometheus / Grafana / OCI	Deployment Budget: █0 (Always Free Tier)
Notification Stack: Telegram Bot / Gmail SMTP / Fast2SMS	Escalation Ladder: 7-Stage Multi-Channel Matrix

1. Executive Summary & Business Objective

Modern enterprise operations require resilient, uninterrupted observability across geographically separated cloud nodes and edge infrastructure. When mission-critical edge gateways or cloud services experience hardware faults, power loss, or network partitions, operations teams require sub-second triage, precise SLA tracking, and multi-channel escalation while strictly eliminating false-positive alarm storms.

This platform delivers an autonomous, 100% open-source observability and ITIL incident escalation framework deployed across Oracle Cloud Infrastructure (OCI) and on-premises edge hardware. It operates 24/7 without requiring personal laptops or local computers to remain running, incurs █0 recurring charges, and provides guaranteed sub-15-second outage detection and sub-10-second recovery notifications.

2. Monitored Infrastructure Targets

Target ID	Device Role	Host & Network Address	Telemetry Protocol	SLA Threshold
Target 1	Secondary Router (TP-Link TL-WR845N v4)	192.168.1.7 MAC: D8:44:89:F5:41:FA (Edge Private LAN)	Outbound UDP 514 Syslog Layer-2 Port 3 Forwarding	Outage: <15s Recovery: <10s
Target 2	Cloud Drive Server (Oracle Cloud VM 1)	161.118.180.68 deepak-cloud-drive.duckdns.org (Public Cloud)	Synthetic HTTPS Blackbox HTTP 200 OK Probing	Outage: <15s Recovery: <10s

3. Network Topology & Out-of-Band CGNAT Traversal

A critical engineering barrier in monitoring on-premises SOHO hardware from the public cloud is **Carrier-Grade NAT (CGNAT)**. Residential and branch broadband connections share dynamic public IP pools (e.g., 103.181.90.226), with all inbound ports blocked by the ISP firewall. Standard cloud pollers are strictly unable to initiate incoming connections to 192.168.1.7.

The Architectural Solution: Instead of inbound polling, the on-premises primary gateway (ZTE F670LV9) is configured to stream remote syslog telemetry **outbound** to Oracle Cloud VM 2 on UDP port 514. Outbound UDP effortlessly traverses CGNAT state tables. The secondary router is hardwired to physical **Port 3 (eth2)** of the gateway's internal bridge (br0). Every physical power

cycle, unplug event, or auto-negotiation reset triggers an immediate sub-second kernel log packet to the cloud.

Key Architectural Benefit: Zero client software or agent daemons are installed on home computers. Telemetry generation is offloaded 100% to the edge router hardware, enabling true 24/7 autonomous monitoring even when all local laptops and workstations are powered off.

4. Corporate 7-Stage Escalation Ladder Specification

The incident watchdog engine (router_watchdog.py) enforces an exact multi-tier escalation ladder measured continuously from the precise onset timestamp of an infrastructure outage:

Ladder Tier	Elapsed Downtime	Severity Level	Telegram & Email Alert Subject	Operational SLA Action
Tier 1	15 seconds	Critical	[1/7 OUTAGE] Immediate Outage Detected	Rapid triage alert dispatched to Telegram, Email, and SMS.
Tier 2	2 minutes (120s)	Major	[2/7 OUTAGE] Outage Confirmed	Confirms sustained outage; eliminates transient boot noise.
Tier 3	10 minutes (600s)	Sustained	[3/7 OUTAGE] Outage Sustained	SLA breach warning; triggers engineering triage.
Tier 4	12 hours (43,200s)	Major Extended	[4/7 OUTAGE] Major Extended Outage	Escalated notification for extended infrastructure loss.
Tier 5	24 hours (86,400s)	Severe Extended	[5/7 OUTAGE] Severe Extended Outage	Daily executive incident status notification.
Tier 6	1 week (604,800s)	Prolonged	[6/7 OUTAGE] Prolonged Outage - 1 Week	Long-term dormant infrastructure operational warning.
Tier 7	1 month (2.59M s)	Critical Dormant	[7/7 OUTAGE] Critical Dormant Host - 1 Mo	Final decommissioning or replacement trigger.

5. Recovery Notification Rules & Debouncing Protocol

To prevent alert storms and operator confusion during restoration, the platform enforces strict recovery rules:

Channel	Dispatch Volume	Format Specification & Footer
Telegram Bot	Strictly 1 Message	Header: ■ [RESOLVED] INFRASTRUCTURE RECOVERED Metrics: Device, Target, Status, Restored Timestamp, Downtime. Footer: All escalation alerts cleared. System operating normally.
Gmail SMTP	Strictly 2 Emails	Email 1: ■ [RESOLVED 1/2] <Target> Back Online (Immediate) Email 2: ■ [RESOLVED 2/2] <Target> Telemetry Restored (+3s delay)
Boot Flap Notice	Transient Debounce	If an Ethernet renegotiation flap occurs during cold boot (<15s), dispatches a clean ■■ [SYSTEM INFO] BOOT LINK STABILIZED notice on Telegram with zero redundant emails.

6. Production Deployment & Systemd Service Architecture

Both core background daemons run on Oracle Cloud VM 2 under native Linux systemd supervisors configured with automatic restart upon failure:

Service Name	Executable Path	Port Bindings	Role & Responsibilities
router-syslog.service	/home/ubuntu/router_syslog_exporter.py	UDP 514 (Syslog) TCP 9125 (Prometheus)	Ingests UDP syslog packets, parses bridge port states, exports Prometheus metrics.
router-watchdog.service	/home/ubuntu/router_watchdog.py	Pulls :9125 & :9090 Direct HTTPS Probes	Multi-target state machine, 7-stage escalation schedule, Telegram/SMTP dispatch.

7. Security, Financial Budget & SLA Compliance

Total Monthly Cost: █0.00 (100% Always Free Tier OCI)	Target 1 Availability SLA: 99.9% Target / Sub-15s Detection
Credential Storage: Restricted 0600 JSON store (alert_config.json)	Target 2 Health Check: Synthetic HTTP 200 OK Probe / 3s Interval
Client Host Impact: 0% CPU / Zero software on user laptops	Repository Status: Synchronized on GitHub main branch

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